

VIVEKAVARDHAN DEVARASETTY

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Professional Summary

Accomplished Automotive Systems Engineer with expertise in delivering Automotive ECUs and a proven track record in analyzing requirements, creating technical specifications, and developing robust test cases. Experienced in assisting with HARA (Hazard Analysis and Risk Assessment) and TARA (Threat Analysis and Risk Assessment) to ensure functional safety and cybersecurity compliance. Demonstrated proficiency in defect management and root cause analysis to guarantee optimal system performance. Experienced in Agile environments, using Jira, Doors and ALM to facilitate effective team collaboration and maintain comprehensive documentation. Specializes in Charging system development, PEPS, infotainment and telematics system validation, driving productivity through innovative problem-solving and effective communication.

Experience

Power Conversion Charging System engineer Ford Motor Company

10/2023 to 12/2024
Dearborn, MI

- Defined and managed system-level requirements for On-Board Charger (OBC), Off-Board Charge Controller (OBCC), and combo modules across BEV and PHEV platforms, ensuring alignment with functional and regulatory standards.
- Collaborated with cross-functional teams, including Design release engineers, software architects to refine and validate requirements, ensuring compliance with GPDS milestones for OBC and OBCC modules.
- Reviewed and maintained interface specifications, defining the sequence diagrams for charging system
- Assisted in HARA and TARA to identify potential safety hazards and cybersecurity threats, incorporating mitigation strategies into system requirements.
- Facilitated requirements validation through test strategy development, supporting system integration and verification teams to ensure software compliance with defined specifications.
- Analyzed telematics data and diagnostic trouble codes (DTCs) using tools like CANoe, Wireshark, and CMS (Connected Mobility Services) to provide insights for requirement updates and issue resolution.
- Maintained DFMEA/DVP&R documentation, ensuring system design robustness and continuous improvement based on warranty data and field feedback.
- Developed and enforced bidirectional traceability between system, software, and validation requirements, ensuring compliance with ASPICE and functional safety guidelines.
- Worked closely with suppliers and internal stakeholders to clarify technical expectations, resolve ambiguities, and ensure seamless integration of new features into vehicle architectures.
- Led requirement change management activities, assessing the impact of software and hardware modifications while ensuring alignment with overall vehicle architecture.
- Authored and reviewed system requirement specification (SRS) and Interface specifications (IFS) to ensure clarity, completeness, and feasibility for implementation.
- Supported end-to-end system verification and validation activities, collaborating with vehicle program teams to develop test scenarios based on system requirements

System Integration Engineer – Vehicle Cybersecurity Epitex onsite at Ford Motor Company

08/2023 to 10/2023

Dearborn, MI

- Led the integration and validation of CAN Message Authentication (CMA) feature for next-generation vehicle platforms from late development to launch.
- Collaborated with cross-functional teams to refine CMA design and ensure seamless integration into the vehicle architecture
- Provided critical technical insights and status updates to stakeholders, enabling informed decision-making on system performance and issue resolution.

- Optimized system implementation workflows, reducing the execution time for Key Distribution script processing and enhancing operational efficiency.
- Identified and addressed potential system-level roadblocks, mitigating risks to ensure timely program deliverables.
- Worked closely with launch teams, troubleshooting emerging system issues and prioritizing critical fixes to maintain project timelines.
- Documented system failures and End-of-Line (EOL) faults, ensuring traceability and supporting root cause analysis for continuous improvement.

Feature Systems Engineer– Public Charging 02/2023 to 07/2023

Epitec onsite at Ford Motor Company

Dearborn, MI

- Developed and defined system-level requirements for public charging features, including Plug and Charge (PnC) and Charge Assist, ensuring seamless functionality and compliance with industry standards.
- Led cross-functional collaboration for vehicle-to-cloud software integration, ensuring alignment with project milestones and technical feasibility.
- Applied structured DFMEA methodologies to proactively identify and mitigate potential system vulnerabilities, enhancing overall system reliability.
- Owned the end-to-end data architecture and integration process, ensuring accurate and secure vehicle-to-cloud communication.
- Defined certificate and authorization handling processes for Plug and Charge, leveraging ISO 15118 and OCPP to ensure secure and efficient authentication.
- Conducted prototype and bench testing, validating CAN data accuracy and system functionality to ensure compliance with requirements.
- Authored and maintained system specifications for End-of-Line (EOL) configuration, providing clear guidelines for production teams.
- Performed in-depth defect analysis, diagnosing root causes and collaborating with ECU and cloud teams to implement corrective actions.

System application engineer – Car Access Systems

08/2022 to 02/2023

Valeo NA

Troy, MI

- Defined and developed system-level requirements for Passive Entry Passive Start (PEPS) ECU, integrating RF, LF to enhance system performance and ensure seamless operation across vehicle platforms.
- Collaborated with OEMs to resolve specification discrepancies, ensuring compliance with remote access, keyless entry, and PEPS functionality requirements while aligning with industry standards.
- Calibrated the RFHM module to optimize system performance for remote start, passive entry, and proximity wakeup, improving user experience and reliability.
- Developed and maintained system architecture documentation, ensuring clear traceability between requirements, design, and validation plans.
- Led root cause analysis for system malfunctions, leveraging advanced debugging tools like CANoe, CDA, and oscilloscope-based signal analysis to diagnose and resolve complex system issues.
- Designed and validated CAN communication protocols, ensuring robust message transmission between BCM RFHM, and other vehicle ECUs.
- Developed and executed system-level test strategies, on bench and Vehicle to validate PEPS ECU performance under various conditions.
- Reviewed wiring schematics and PIN-out diagrams, ensuring correct signal routing, proper grounding, and efficient power distribution for RFHM modules.
- Worked closely with hardware and electrical teams to define ECU connector layouts, verify signal integrity, and troubleshoot wiring-related issues.
- Coordinated with manufacturing and validation teams to define End-of-Line (EOL) testing procedures, ensuring PEPS ECU meets production standards.
- Facilitated seamless communication between OEMs and Valeo engineering teams, ensuring alignment on technical specifications, design changes, and issue resolution.

Infotainment and Telematics Test Engineer**03/2020 to 07/2022****PROLIM Global Corporation onsite at Toyota North America****Plano, TX**

- Responsible for complete end-to-end testing for telematics and infotainment features, which include vehicle-level testing, mobile app testing, OTA, connected audio, navigation, and search.
- Ensure full test coverage of all implemented internal requirements before each sprint cycle.
- Customized and executed OTA campaigns to support diverse ECU functionalities.
- Configure and Setup Test Benches for different vehicle variants
- Conducted testing and validation of Toyota app telematics features.
- Conducted black box, system, end-user, regression, bug fix, and end-to-end testing.
- Ensured accurate reporting of issues by leveraging JIRA alongside comprehensive documentation.
- Collaboratively worked with suppliers and cloud service teams to resolve problems.
- Executed detailed analysis of HU logs with support from Postman, gRPC UI, and Datadog to confirm accuracy of API interactions.

Test engineer – Infotainment**07/2017 to 03/2020****Open Systems Inc. onsite at Panasonic Automotive****The Colony, TX**

- Collaborated with development and systems engineering teams to define features and capabilities.
- Developed test cases based on the determined functional and performance criteria of modules.
- Participated in the assessment of various infotainment functionalities like BT, AM/FM and Navigation.
- Engaged in factory evaluations alongside clients for quality assurance.
- Designed and managed System Validation Plans (DVP) to ensure product compliance.
- Maintain test environments through re-flashing of latest software releases.
- Created traceability matrices between user stories and test cases.

Software Analyst**01/2014 to 07/2015****Accenture****Bangalore**

- Coordinating with the customer to gather requirements and develop their requirements in SAP
- As needed, performed functional design reviews and gap analysis
- Applying project management skills by guiding the team to a timely delivery of projects
- Conducting technical reviews and evaluations to test the functionality and investigate the errors
- Good understanding of Software Development process (V-Model) and Bug life cycle

Education**Masters: Automotive Systems Engineering****01/2017**

University of Michigan

Dearborn

Bachelors: Mechanical Engineering**01/2013**

Sri Krishnadevaraya University

Anantapur, India

Technology summary

Wireshark, PuTTY, CDA (Chrysler), DET (Ford), NFC, BLE, ISO 15118-2, OCPP, SAE J1772, CAN, LIN, CAN-FD, CANoe, CANalyzer, Vehicle Spy, Jira, IBM RTC, Redmine, Team Center, EOL tools, Federal safety standards, DFMEA, ISO 26262, ISO 21434, ASPICE.